Entropy-Stable Multirate Time-Integration through Paired-Explicit Relaxation Runge-Kutta Methods

ICOSAHOM 2025

MS 132 - Advances in provably stable high-order discretizations of non-linear PDEs and their applications

Daniel Doehring[1]

in collaboration with

H. Ranocha[2] and M. Torrilhon[1]

[1] **ACoM, RWTH Aachen University** [2] IoM, Johannes Gutenberg University Mainz

2025/07/15

Outline

Entropy-Stable Multirate Time-Integration through Paired-Explicit Relaxation Runge-Kutta Methods

1. Paired-Explicit Runge-Kutta (P-ERK)
2. Relaxation Runge-Kutta (RRK)
3. Validation
4. Applications
5. Conclusions

Partitioned Runge-Kutta Methods: Scope

Consider PDEs of the form

$$\partial_t \mathbf{u}(t, \mathbf{x}) + \nabla \cdot \mathbf{f}\big(\mathbf{u}, \nabla \mathbf{u}\big) + \mathbf{g}(\mathbf{u}, \nabla \mathbf{u}) = \mathbf{s}\big(t, \mathbf{x}; \mathbf{u}\big)$$

which describe e.g. compressible flows and related systems.

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We employ a method-of-lines approach (via the DGSEM) to obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{U}(t) = \mathbf{F}\big(t, \mathbf{U}(t)\big)$$

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$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{U}(t) = \mathbf{F}\big(t, \mathbf{U}(t)\big)$$

and target **convection-dominated** PDEs, i.e.,

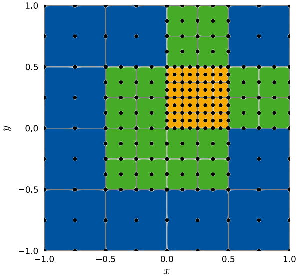
$$\rho \sim \frac{|a|}{\Delta x} \gg \frac{d}{\Delta x^2}$$

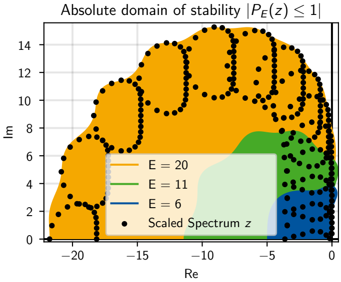
where ρ is the spectral radius of

$$J_{\mathbf{F}}(\mathbf{U}) := \frac{\partial \mathbf{F}}{\partial \mathbf{U}}\bigg|_{\mathbf{U}}.$$

Partitioned Runge-Kutta Methods

Multirate time-integration by using **stabilized/optimized** schemes in regions with higher characteristic speeds (stricter CFL)





- For *convection-dominated* problems:

$$\Delta t \overset{!}{\leq} C_t(E) \cdot C_x(k) \min_{i,j} \frac{\Delta x_i}{|\mu_j(\mathbf{u}_i)|}$$

- For *optimized* Runge-Kutta schemes:

$$\Delta t_{p;E} \sim C_t(E) \sim E$$

Partitioned Runge-Kutta Methods

Partition ODE according to characteristic speeds $a_i := \min_j \frac{\Delta x_i}{|\mu_j(\mathbf{u}_i)|}$

$$\mathbf{U}'(t) = \begin{pmatrix} \mathbf{U}^{(1)}(t) \\ \vdots \\ \mathbf{U}^{(R)}(t) \end{pmatrix}' = \begin{pmatrix} \mathbf{F}^{(1)}\left(t, \mathbf{U}^{(1)}(t), \dots, \mathbf{U}^{(R)}(t)\right) \\ \vdots \\ \mathbf{F}^{(R)}\left(t, \mathbf{U}^{(1)}(t), \dots, \mathbf{U}^{(R)}(t)\right) \end{pmatrix} = \mathbf{F}\left(t, \mathbf{U}(t)\right)$$

and employ different RKMs

$$\mathbf{K}_i^{(r)} = \mathbf{F}^{(r)}\left(t_n + c_i \Delta t, \mathbf{U}_n + \Delta t \sum_{j=1}^{S} \sum_{k=1}^{R} a_{i,j}^{(k)} \mathbf{K}_j^{(k)}\right),$$

$$\mathbf{U}_{n+1} = \mathbf{U}_n + \Delta t \sum_{i=1}^{S} b_i \sum_{k=1}^{R} \mathbf{K}_i^{(r)}$$

optimized for the spectrum $\boldsymbol{\sigma}\left(\frac{\partial \mathbf{F}}{\partial \mathbf{U}}\Big|_{\mathbf{U}_0}\right)$.

Paired-Explicit Runge-Kutta Methods

- $S = 6,\ p = 2$ **Paired**[1] $E = \left\{E^{(1)}, E^{(2)}\right\} = \{3, 6\}$ Runge-Kutta

i	$\mathbf{c}$	$A^{(1)}$						$A^{(2)}$					
1	0												
2	$1/10$	$1/10$						$1/10$					
3	$2/10$	$2/10$	0					$2/10 - a_{3,2}^{(2)}$	$a_{3,2}^{(2)}$				
4	$3/10$	$3/10$	0	0				$3/10 - a_{4,3}^{(2)}$	0	$a_{4,3}^{(2)}$			
5	$4/10$	$4/10$	0	0	0			$4/10 - a_{5,4}^{(2)}$	0	0	$a_{5,4}^{(2)}$		
6	$5/10$	$5/10 - a_{6,5}^{(1)}$	0	0	0	$a_{6,5}^{(1)}$		$5/10 - a_{6,4}^{(2)}$	0	0	0	$a_{6,4}^{(2)}$	
	$\mathbf{b}^T$	0	0	0	0	0	1	0	0	0	0	0	1

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- Method $r = 1$ requires $\mathbf{F}^{(1)}$ *evaluation only for* $i = \mathbf{1, 5, 6}$

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- Method $r = 1$ requires $\mathbf{F}^{(1)}$ *evaluation only for* $i = \mathbf{1}, \mathbf{5}, \mathbf{6}$
- Method $r = 2$ requires $\mathbf{F}^{(2)}$ *evaluation for all* $i = \mathbf{1}, \dots, \mathbf{6}$

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Advantages:

- Low-Storage: Only two stage registers $\mathbf{K}_1, \mathbf{K}_i$

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Issue:

- *Prov.* *not* Strong Stability Preserving (SSP) since $\exists i : b_i = 0$[1]

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Nonlinear Stability of Time-Integration Schemes

- **Strong-Stability Preserving (SSP)**: TVD/TVB Preservation

Nonlinear Stability of Time-Integration Schemes

- **(Modified) Patankar Runge-Kutta (MPRK)**:
  Unconditional Positivity Preservation

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Nonlinear Stability of Time-Integration Schemes

Robustness/Nonlin. Stab.

- **(Modified) Patankar Runge-Kutta (MPRK)**:
  Unconditional Positivity Preservation
- **Strong-Stability Preserving (SSP)**:
  TVD/TVB Preservation
- **Relaxation Runge-Kutta (R-RK)**:
  Convex Functional (Entropy) Preservation

Flexibility

Notion of Entropy η

Mathematical entropy $s(\mathbf{u})$ of an ideal gas with $\mathbf{u} = (\rho, \rho\mathbf{v}, \rho e)$

$$s(\mathbf{u}) := -\rho \cdot s_{\text{therm}}(\mathbf{u}) = -\rho \cdot \ln\left(\frac{p(\mathbf{u})}{\rho^\gamma}\right)$$

Total/Global Entropy η

$$\eta(t) := \eta\big(t; \mathbf{u}(t, \mathbf{x})\big) = \int_\Omega s\big(\mathbf{u}(t, \mathbf{x})\big)\,\mathrm{d}\mathbf{x}\,.$$

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Entropy–conservative semidiscretizations $\mathbf{U}' = \mathbf{F}(\mathbf{U})$ satisfy

$$0 = \frac{\mathrm{d}}{\mathrm{d}t}\eta\big(\mathbf{U}(t)\big) = \left\langle \frac{\partial \eta(\mathbf{U})}{\partial \mathbf{U}}, \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{U}(t) \right\rangle = \left\langle \frac{\partial \eta(\mathbf{U})}{\partial \mathbf{U}}, \mathbf{F}\big(t, \mathbf{U}(t)\big) \right\rangle$$

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Entropy: A non-linear Functional

"True" entropy evolution: Application of RKM to

$$\eta'(t) = \langle \partial_{\mathbf{U}} \eta, \mathbf{F}(\mathbf{U}) \rangle$$

yields

$$\eta_{n+1} = \eta_n + \Delta t \sum_{i=1}^{S} b_i \left\langle \frac{\partial \eta(\mathbf{U}_{n,i})}{\partial \mathbf{U}}, \mathbf{F}(\mathbf{U}_{n,i}) \right\rangle$$

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But: We evolve $\mathbf{U}' = \mathbf{F}(\mathbf{U})$!

$$\Rightarrow \eta(\mathbf{U}_{n+1}) = \eta \left(\mathbf{U}_n + \Delta t \sum_{i=1}^{S} b_i \mathbf{F}(\mathbf{U}_{n,i}) \right)$$

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$$\Rightarrow \eta(\mathbf{U}_{n+1}) = \eta\left(\mathbf{U}_n + \Delta t \sum_{i=1}^{S} b_i \mathbf{F}(\mathbf{U}_{n,i})\right) \neq \eta_{n+1}$$

The Relaxation Parameter γ

Introducing $\gamma \in \mathbb{R}$ allows[1,2]

$$\eta\left(\mathbf{U}_{n+1}(\gamma_n)\right) \overset{!}{=} \eta(\mathbf{U}_n) + \gamma_n \Delta t \sum_{i=1}^{S} b_i \left\langle \frac{\partial \eta(\mathbf{U}_{n,i})}{\partial \mathbf{U}_{n,i}}, \mathbf{F}(\mathbf{U}_{n,i}) \right\rangle$$

$$\eta\left(\mathbf{U}_n + \gamma_n \Delta t \mathbf{b}^T \mathbf{K}\right) \overset{!}{=} \eta(\mathbf{U}_n) + \gamma_n \Delta t \sum_{i=1}^{S} b_i \langle \mathbf{W}_{n,i}, \mathbf{F}(\mathbf{U}_{n,i}) \rangle \tag{1}$$

proven for general linear methods (such as PRKMs/P-ERK)[3]

[1] Ketcheson; SIAM J. Numer. Anal.; 2019

[2] Ranocha et al.; SIAM J. Numer. Anal.; 2020

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Solve (1) for γ_n at each time step

$\Rightarrow$ Newton; Convergence to (near) machine precision within $\sim$ 3 steps

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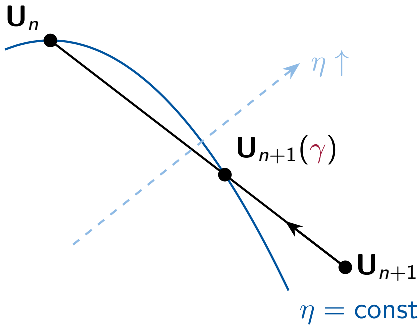
Illustration: Entropy-Conservative Problem

$$\mathbf{U}_{n+1}(\gamma) = \mathbf{U}_n + \gamma \Delta t \mathbf{b}^T \mathbf{K}$$
$$\mathbf{U}_{n+1} \equiv \mathbf{U}_{n+1}(1)$$

Anti-Dissipative Step:



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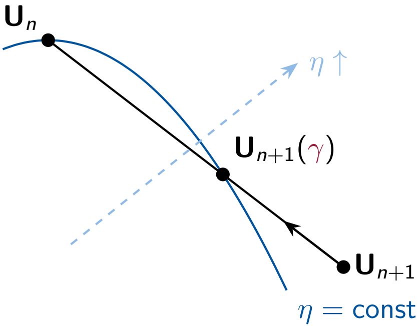
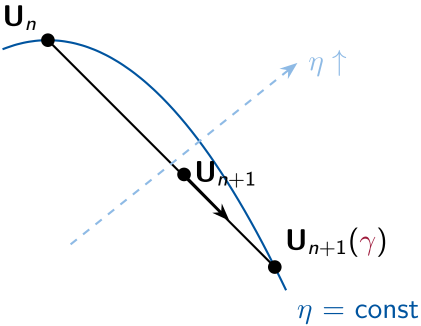
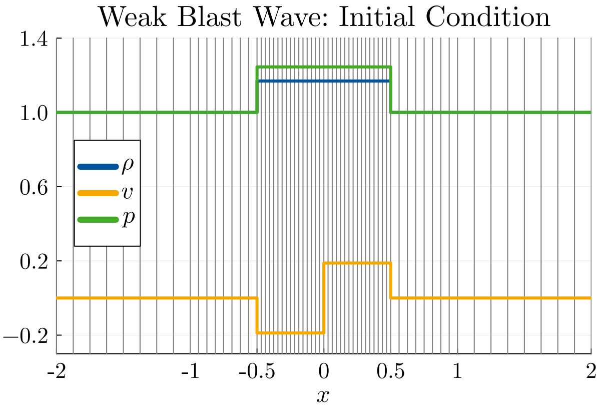


Dissipative Step:



Validation: Entropy Preservation

- 1D Euler, Ranocha-Flux, DGSEM w/ Flux-Differencing, $k = 3$
- Weak Blast Wave[1] on 3-level mesh/P-ERRK methods



[1] Hennemann et al.; JCP; 2021

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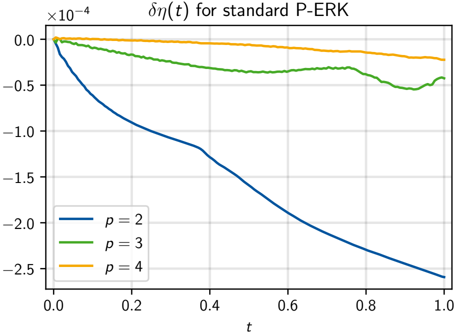
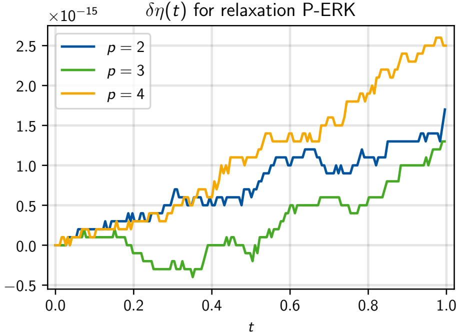
$$\delta\eta(t) := \eta(t) - \eta(t_0)$$





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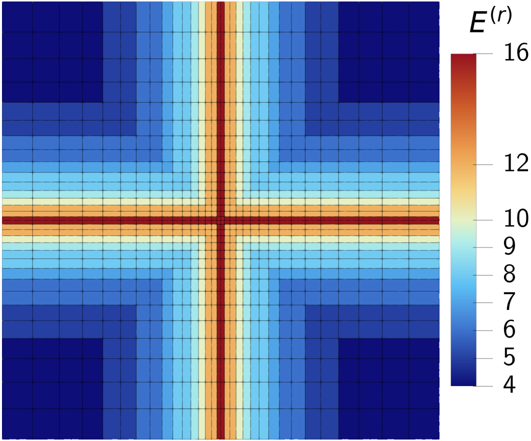
Validation: Entropy Preservation

- 2D Euler, Ranocha-Flux, DGSEM w/ Flux-Differencing, $k = 3$
- Isentropic Vortex on 32^2 cells
- "Continuous" refinement, 9 P-ERRK levels



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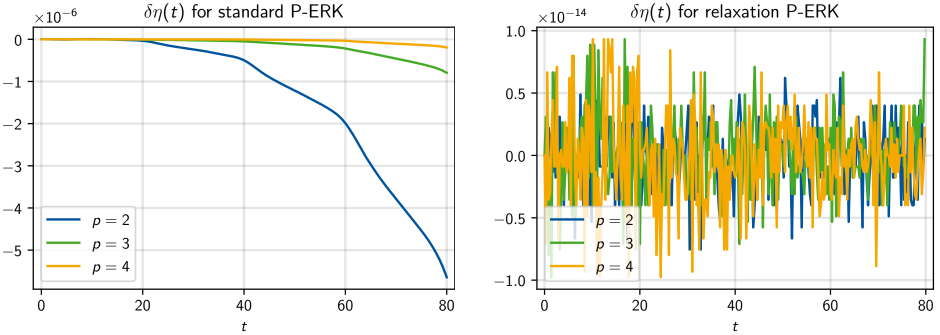
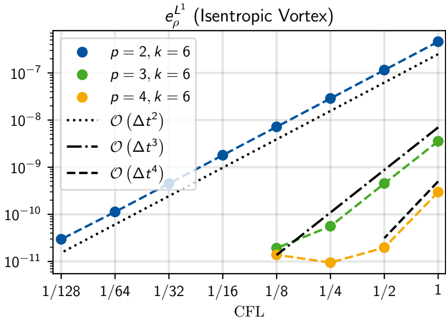
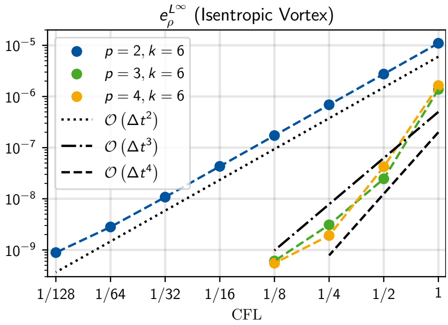


Validation: Convergence

- 2D Euler, HLLC, weak-form DGSEM, $k = 6$
- Isentropic Vortex Advection, 64^2 base discretization
- 3-level adaptively refined mesh

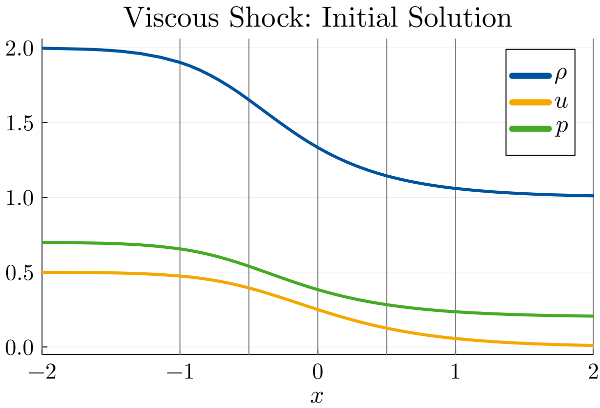
$$e_\rho^{L^1}(t) := \frac{1}{|\Omega|} \int_\Omega \left| \rho(t, \mathbf{x}) - \rho_{\text{exact}}(t, \mathbf{x}) \right| \mathrm{d}\mathbf{x}$$





Validation: Convergence

- 1D Navier-Stokes, HLLE, weak-form DGSEM, BR1, $k = p - 1$
- Viscous shock: Becker-Morduchow-Libby solution
- Diffusion dominated: $\Delta t \sim \Delta x^2 \Rightarrow$ 2-Level mesh/P-ERRK methods



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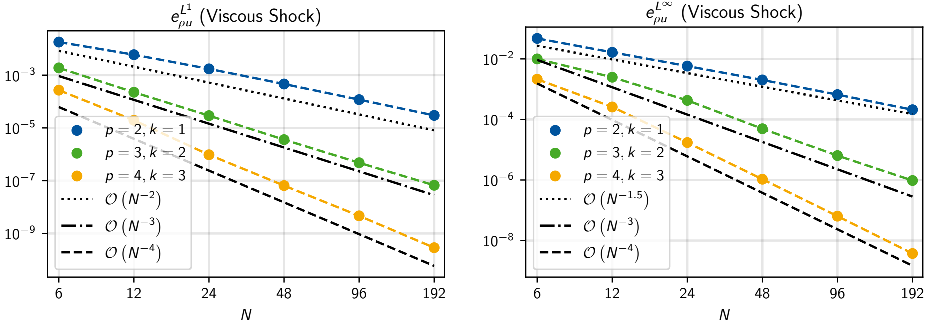
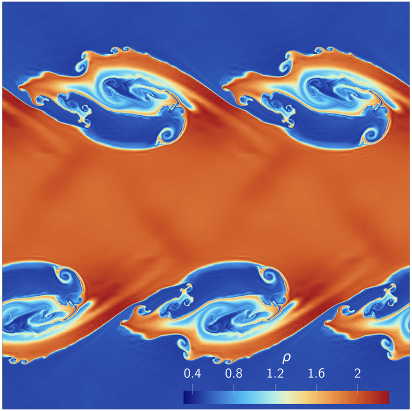
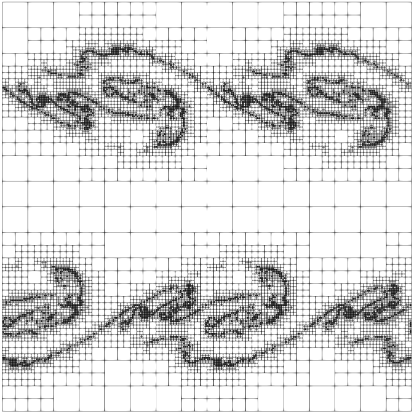


Applications: Under-resolved simulations

Kelvin-Helmholtz Instability

- $k = 3$, HLLE, Subcell-Limiting & Flux-Differencing
- 6-level mesh, 3-level P-ERRK scheme, $t_f = 3.2$





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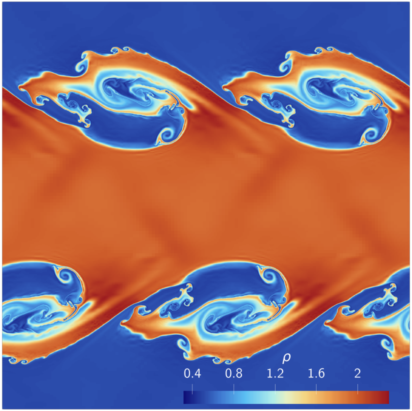
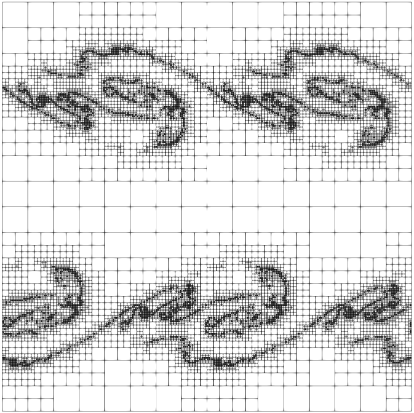
- Non-relaxed scheme crashes at $t \sim 2.73$! $N|_t = 1267$ vs $N_\gamma|_t = 1184$

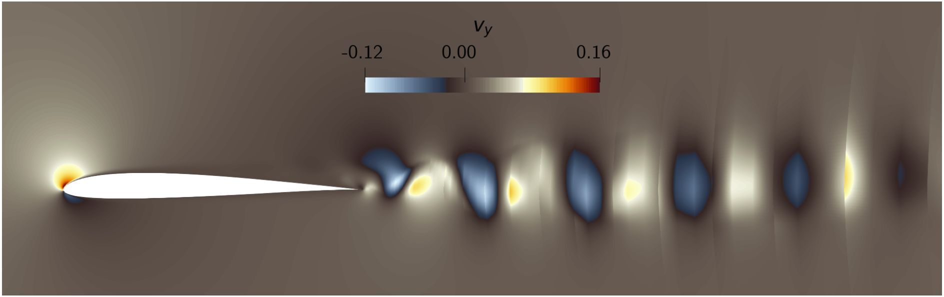




Applications: Under-resolved simulations

SD7003 Airfoil

- Ma $= 0.2$, Re $= 10^4$, Pr $= 0.72$, $\alpha = 4^\circ$
- $k = 3$, HLLC, Flux-Differencing with Ranocha-Flux, BR1
- $\Omega = [-20, 40] \times [-20, 20]$, 7605 linear quads[1]



[1] Vermeire; JCP; 2019

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Source	$\overline{C}_L$	$\overline{C}_D$
P-ERRK $p = 4$	0.3827	0.04995
P-ERK $p = 2$[1]	0.3841	0.04990
P-ERK $p = 3$[2]	0.3848	0.04910
Uranga et al.[3]	0.3755	0.04978
López-Morales et al.[4]	0.3719	0.04940

[1] Vermeire; JCP; 2019

[2] Nasab, Vermeire; JCP; 2022

[3] Uranga et al.; Int. J. Num Methods Eng.; 2011

[4] López-Morales et al.; 32nd AIAA AAC.; 2014

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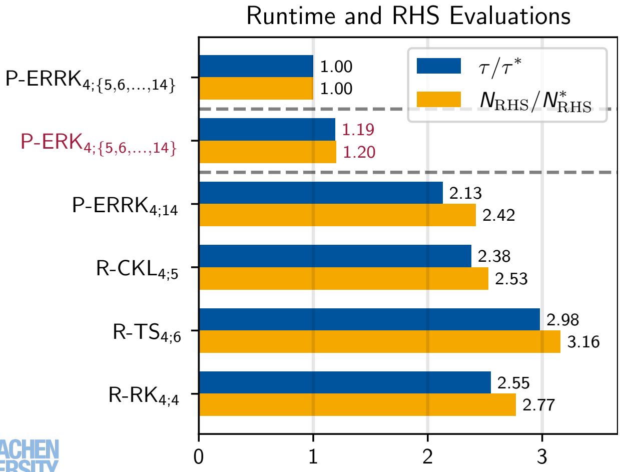
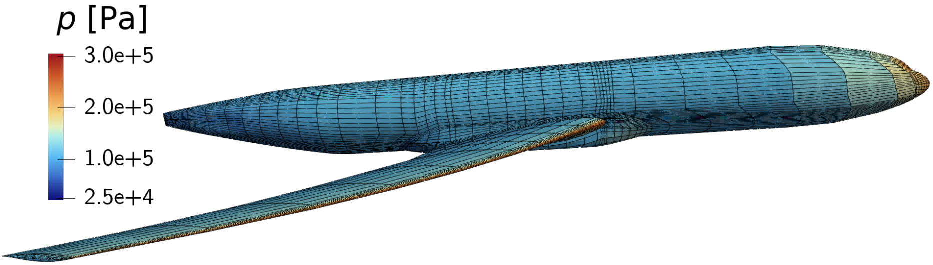


Applications: Under-resolved simulations

Common Research Model (CRM)

- $\mathrm{Ma} = 0.85$, $\mathrm{Re} = 200 \cdot 10^6$, $\mathrm{Pr} = 0.72$
- $k = 2$, HLL, Flux-Differencing with Ranocha-Flux, BR1
- 79505 linear hexahedra[1]



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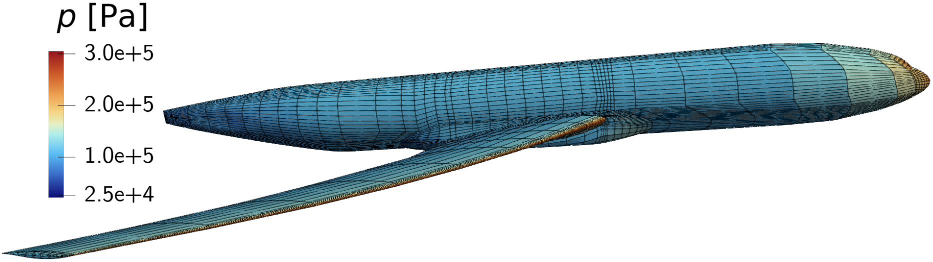

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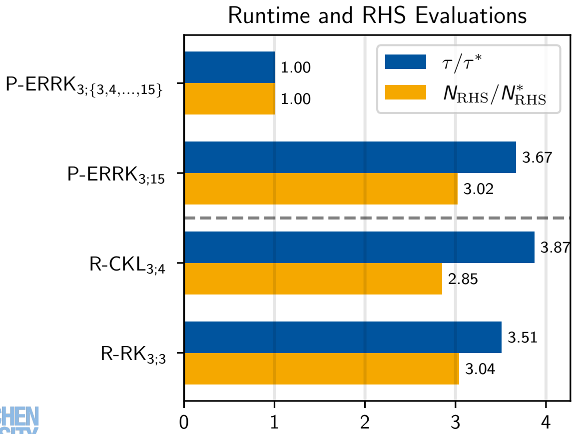
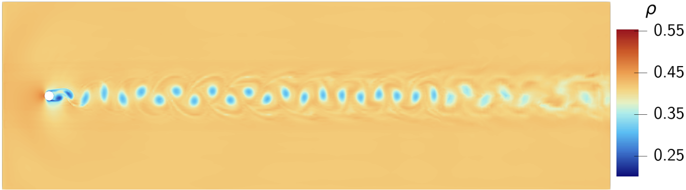


Applications: Well-resolved simulations

Visco-Resistive Flow around a Cylinder

- Magnetized flow around cylinder with $B = 0$[1]
- $\mathrm{Ma} = 0.5$, $\mathrm{Pr} = 0.72$, $\mathrm{Re} = 200$, $S_\mu = S_\nu = 50$

Navier-Stokes-Fourier



Visco-Resistive MHD

[1] Warburton, Karniadakis; JCP; 1999

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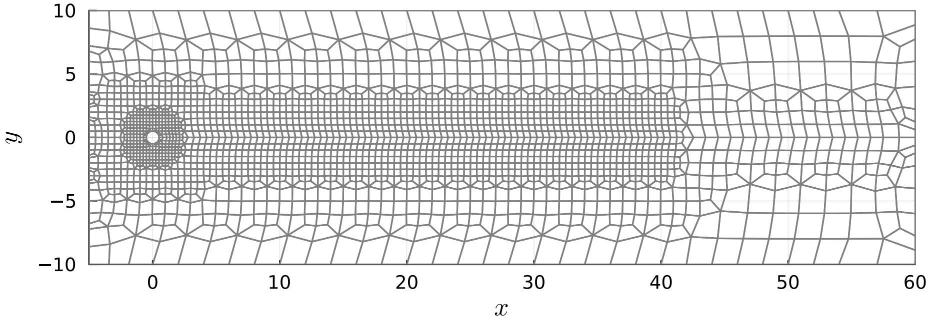

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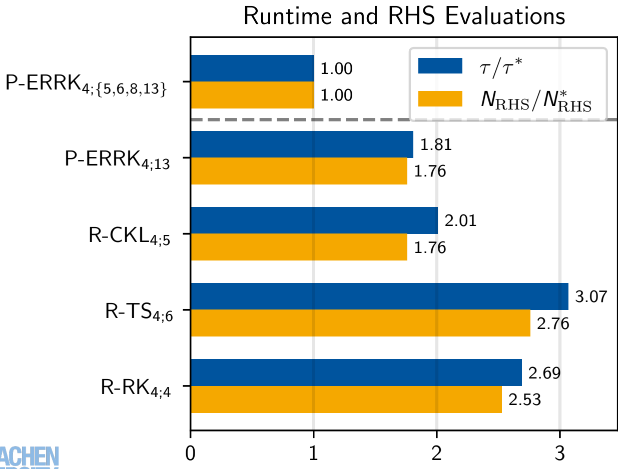
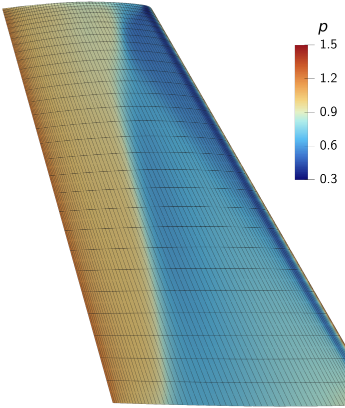


Applications: Well-resolved simulations

Inviscid ONERA M6 Wing

- $\mathrm{Ma} = 0.84$, $\alpha = 3.06°$, 294838 hexahedra[1]
- $k = 2$, Rusanov, Flux-Diff. (no subcell limiting needed!)



Lift-Coefficient

$$C_L := \frac{L(p)}{0.5\rho_\infty U_\infty^2 A}$$

Trixi.jl, Projected A: $C_L \approx 0.2951$
SU2, Computed A: $C_L \approx 0.2861$ (-3%)

[1] Slater; NASA NPARC Validation Archive; 2002

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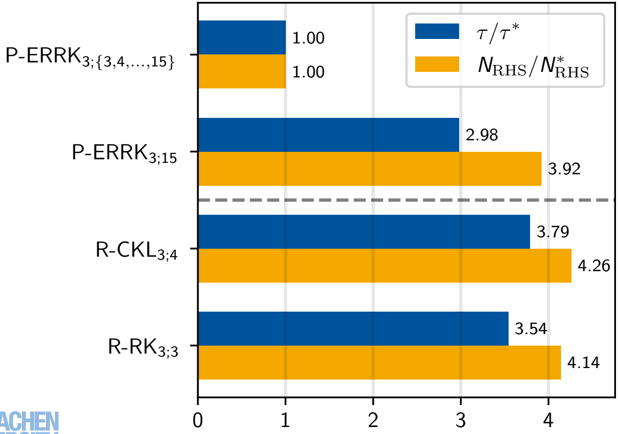
Runtime and RHS Evaluations



Conclusions

Paired-Explicit Relaxation Runge-Kutta (P-ERKK) Methods

- *Multirate Method*: Maximum Speedup $\mathcal{O}\left(\frac{\max_r\left\{E^{(r)}\right\}}{\min_r\left\{E^{(r)}\right\}}\right)$
- *Problem–Optimized*: Maximal linear stability
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Observations:

- Few Newton iterations to solve relaxation equation
- Larger timesteps and/or longer simulations for under-resolved simulations
- No recorded benefit for well-resolved simulations
- Global entropy η computation relatively expensive

Thank You for your attention!

Questions?

doehring@acom.rwth-aachen.de
https://www.acom.rwth-aachen.de
https://github.com/DanielDoehring
https://doi.org/10.48550/arXiv.2507.04991

Reproducibility Repository: https://github.com/DanielDoehring/paper-2025-perrk

Trixi.jl: https://github.com/trixi-framework/Trixi.jl